

## Top Company Interview Questions – Time Complexity

Disclaimer: When compiling related questions from each lesson, note that not all of them will have answers explicitly mentioned in the material. For such questions, you are encouraged to use ChatGPT and other credible references to find and understand the answers.

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1. What is Time Complexity, and why do we represent it using a mathematical function instead of actual time? (TCS, 2023)
2. Why is measuring program execution in seconds not reliable across different machines or environments? (Infosys, 2021)
3. Explain the analogy between Time Complexity and standardized SI units like the 1 kg weight. (Wipro, 2022)
4. Define the steps involved in determining an algorithm's time complexity. (Accenture, 2023)
5. What is the "basic operation" in time complexity analysis, and why is it important? (Capgemini, 2024)
6. Why do we keep only the *dominant term* while analyzing the time complexity function  $f(n)$ ? (Cognizant, 2023)
7. Differentiate between  $O(1)$ ,  $O(n)$ , and  $O(n^2)$  with simple examples. (Infosys, 2024)
8. Why is array element access considered  $O(1)$ , while traversal is  $O(n)$ ? (TCS, 2022)
9. What does the "Big O" in Big O notation stand for, and what does it represent? (Wipro, 2021)
10. How does time complexity help us choose the right data structure for CRUD operations? (Capgemini, 2023)



11. How does an increase in input size ( $n$ ) affect algorithms of different time complexities? (Amazon, 2022)
12. Compare the time complexities of array insertion and linked-list insertion. (Microsoft, 2024)
13. Why is it important to understand time complexity before writing production-level code? (Infosys, 2023)
14. What is the difference between time complexity and space complexity, and when do we consider both? (Accenture, 2024)
15. How does time complexity provide a fair way to compare two algorithms executed on different hardware? (Google, 2022)

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